

Topic 1 summary: One dimensional kinematics

- Key definitions: position, velocity and acceleration
- Motion under constant (uniform) acceleration
- Motion with variable acceleration
- Velocity and Acceleration as a function of displacement

Key Definitions

1. **Distance**: How far you have travelled regardless of direction. (unit: m).
2. **Displacement**: It is the distance measured (covered) in a particular direction denoted Δx . (unit: m)
3. **Average Velocity**: Rate of change of displacement with time (unit: m/s).

$$\text{avg. velocity} = \frac{\Delta x}{\Delta t}$$

4. **Average speed**: $\frac{\text{distance covered}}{\text{time taken}}$ (m/s).

5. **Instantaneous velocity** ($\vec{v}$).

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{x_{t+\Delta t} - x_t}{\Delta t} = \frac{dx}{dt}$$

It is the first derivative of displacement (x) with time

$$\vec{v} = \frac{dx}{dt} \quad (\text{m/s}).$$

- Note**:- Graphically, dx/dt is the slope of a displacement-time graph at a particular instant in time.
- The instantaneous velocity is zero at the turning points on the graph.

6. **Average acceleration**: rate of change of velocity with time.

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} \quad (\text{m/s}^2).$$

7. **Instantaneous acceleration**:

It is the slope of a velocity-time graph at a particular instant in time.

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\vec{v}_{t+\Delta t} - \vec{v}_t}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2x}{dt^2}.$$

i.e. $\boxed{\vec{a} = \frac{dv}{dt} = \frac{d^2x}{dt^2}} \quad (\text{m/s}^2).$

Motion Under Constant acceleration

Consider a body moving in a straight line with constant or uniform acceleration for a certain time t . The quantities associated with this motion are:-

Initial velocity u

Final velocity v

Time t

Acceleration a .

Total displacement from starting point, x .

Then we have the following standard equations:

$$\begin{aligned} v &= u + at \\ x &= ut + \frac{1}{2}at^2 \\ v^2 &= u^2 + 2ax \end{aligned}$$

Sometimes, velocity-time graphs are used in conjunction with the above equations to solve kinematics problems. The gradient of a velocity-time graph gives the acceleration while the area between the graph and the time axis gives the displacement.

Motion with Variable Acceleration

From calculus, given the displacement, x , as a function of time,

$$\vec{v} = \frac{dx}{dt} \quad \text{and} \quad \vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2x}{dt^2}.$$

Similarly, given the acceleration, $\vec{a}$,

$$\vec{v} = \int \vec{a} dt \quad \text{and} \quad \vec{x} = \int \vec{v} dt.$$

Note: When the instantaneous velocity $\vec{v} = \frac{dx}{dt} = 0$, we achieve maximum/minimum displacement.

When instantaneous acceleration $\vec{a} = \frac{dv}{dt} = 0$, we achieve maximum/minimum velocity.

Velocity and acceleration as a function of displacement.

The velocity or acceleration of a particle is sometimes given as a function of displacement, x , rather than time, t .

Using the chain rule of differentiation

$$\vec{a} = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = \frac{dv}{dx} v$$

$$\text{i.e. } \vec{a} = v \frac{dv}{dx}.$$

But $\frac{d}{dx}(v^2) = 2v \frac{dv}{dx}$ by implicit differentiation

$$\Rightarrow v \frac{dv}{dx} = \frac{1}{2} \frac{d}{dx}(v^2).$$

$$\therefore \vec{a} = v \frac{dv}{dx} = \frac{1}{2} \frac{d}{dx}(v^2)$$

When acceleration $\bar{a}$ is given as a function of displacement, $\bar{v}$ is still found by using $\bar{a} = \bar{v} \frac{d\bar{v}}{dx}$

i.e. given $\bar{a} = f(x)$
we write $\bar{v} \frac{d\bar{v}}{dx} = f(x)$

$$\bar{v} d\bar{v} = f(x) dx$$

then integrate both sides

$$\int \bar{v} d\bar{v} = \int f(x) dx$$

$$\frac{\bar{v}^2}{2} = \int f(x) dx.$$

e.g. If $a = 4x + 2$ and initially $x = 0$ and $v = 1 \text{ m/s}$,
find the value of v when $x = 3 \text{ m}$.

$$\bar{a} = 4x + 2$$

$$\bar{v} \frac{d\bar{v}}{dx} = 4x + 2$$

$$\int \bar{v} d\bar{v} = \int (4x + 2) dx$$

$$\frac{\bar{v}^2}{2} = 2x^2 + 2x + C$$

but $x = 0$ when $v = 1$

$$\therefore \frac{1}{2} = 0 + 0 + C \Rightarrow C = \frac{1}{2}$$

$$\therefore \frac{\bar{v}^2}{2} = 2x^2 + 2x + \frac{1}{2}$$

$$\text{or } \bar{v}^2 = 4x^2 + 4x + 1$$

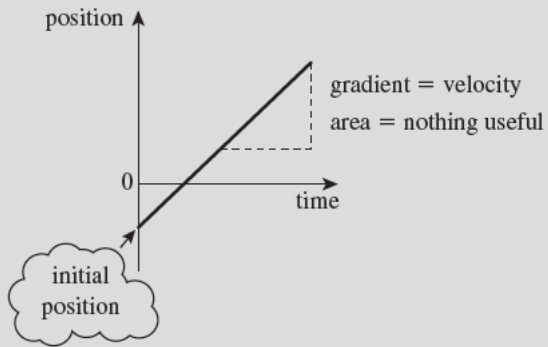
$$\bar{v}^2 = (2x + 1)^2$$

$$\text{or } \bar{v} = (2x + 1)$$

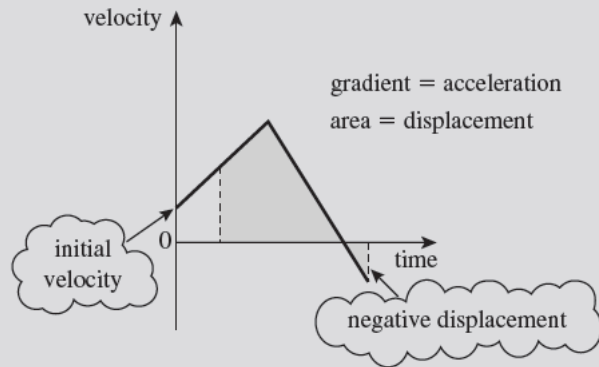
$$\therefore \text{ at } x = 3 \text{ m, } \bar{v} = (2(3) + 1) = \underline{\underline{7 \text{ m/s}}}$$

1.4 Graphs

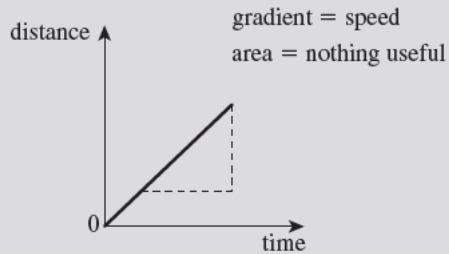
● *Position–time*



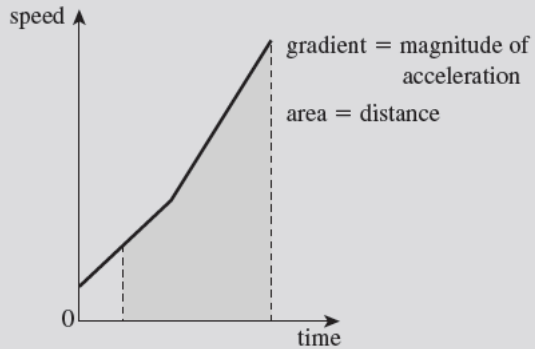
● *Velocity–time*



● *Distance–time*



● *Speed–time*



Vertical motion under gravity

- The acceleration due to gravity ($g \text{ ms}^{-2}$) is 9.8 ms^{-2} vertically downwards.
- Always draw a diagram and decide in advance where your origin is and which way is positive.
- Make sure that your units are compatible.